

Sprint 7: Cached Network Fee Engine

Fast withdrawal fee quotes from fresh network-scoped evidence

Implementation complete; final PostgreSQL rerun and production migration pending

Honest status: ADR-012 is accepted. Domain calculations, schema, migration 12, cache repository, leased worker, strict API, and automated tests are implemented. Migration 12 applied successfully to `sle_test`. The first full Railway run passed 13 of 14 tests; one query after a deliberate concurrent-client race timed out through the proxy. That assertion was reordered and awaits a clean full rerun.

Connection to Earlier Sprints

Sprint 3 provides immutable conversion-rate evidence. Sprint 5 provides network-scoped treasury configuration. Sprint 6 reserves purchased inventory. Sprint 7 adds a separate cache for the blockchain cost of external withdrawals; it does not change purchase quote spreads or fees.

What Was Built

The pure domain engine selects the conservative estimate, checks provider/RPC deviation, applies fixed and percentage buffers with upward integer rounding, and converts native gas cost into the withdrawn asset using immutable rational evidence. PostgreSQL stores versioned policies, observations, conversion evidence, accepted or rejected snapshots, exact snapshot inputs, leased refresh jobs, and immutable withdrawal fee quotes. The worker uses deterministic sandbox adapters behind domain-owned interfaces. The API requires an explicit asset-network and transfer type.

End-to-End Example

```
fee provider estimate ---- independent RPC estimate
      \                               /
      freshness + deviation gate
          |
      immutable cached snapshot
          |
      native gas + buffer + conversion evidence
          |
Sendaza fee request: asset-network + type + amount
          |
      principal + network charge + service fee = total debit
          |
      immutable quote with a short expiry
```

For a token withdrawal, the recipient still receives the exact principal. The treasury may pay gas in the network's native asset, while the customer network-fee charge is converted into the withdrawn asset. USDT on two networks therefore uses two independent policies and caches.

System Boundary

Sendaza owns the customer screen, authorization, and ledger lock. SLE owns the estimate, buffer, conversion evidence, and quote. Sprint 8 will consume the fee quote during Fireblocks submission. It may reject or review a withdrawal if current cost exceeds tolerance, but it cannot silently increase the approved customer debit.

Rules Protected

- Amounts use atomic-unit `bigint` ; API amounts are decimal strings.
- Fee routes are keyed by asset-network and transfer type, never asset alone.
- Stale, missing, duplicated-source, or deviating evidence cannot publish an accepted snapshot.
- Buffers, conversions, and service percentages round upward.
- Recipient principal is unchanged; fees are explicit add-ons.
- Cross-asset customer debits are rejected for the MVP.
- Evidence, snapshots, and quotes are immutable in PostgreSQL.
- Quote creation is idempotency- and audit-protected.

Test and Environment Evidence

Local gates currently cover exact arithmetic, evidence failures, strict request parsing, response precision, and controller behavior. Migration 12 applied to the isolated Railway `sle_test` database and Prisma reported all 12 migrations current. In the first full PostgreSQL run, 13 tests passed and the final post-concurrency query exceeded the proxy timeout. The stale-cache assertion was moved before the concurrency race so it no longer depends on that connection state; the complete rerun remains pending.

Limitations and Next Dependency

The active worker uses deterministic sandbox estimators. Real provider and independent RPC adapters require network-specific integration and credentials before mainnet. Sprint 7 does not submit withdrawals or record mined fees. Production migration 12 requires separate approval after the full isolated suite passes.

Glossary

Native fee asset: the coin a blockchain uses for gas. **Charge asset:** the asset used to recover the network cost from the customer. **Buffer:** an explicit safety amount above the estimate. **Deviation:** the difference between independent estimates. **Snapshot:** immutable cached evidence valid until a fixed time. **Rational**

conversion: an exact numerator and denominator rather than floating-point math. **Lease:** a time-limited worker claim safe to recover after a crash.

Sprint 7 handover report. Canonical behavior lives in ADR-012, API_SPEC.md, ARCHITECTURE.md, and modules/network-fees/MODULE.md.